CURRENT ISSUES
5TH-8TH GRADE STUDY PACKET

OUTDOOR SAFETY & SURVIVAL

Envirothon 2022

Part 1:

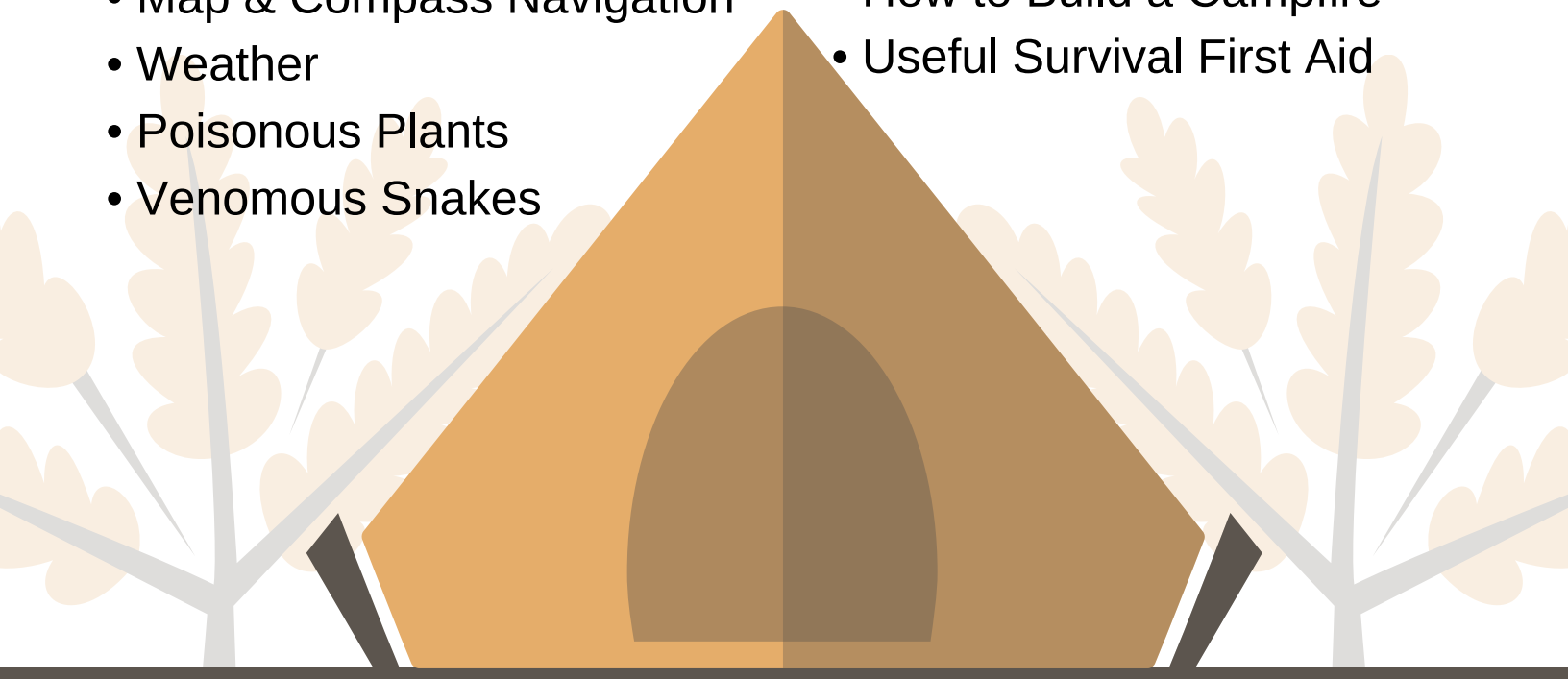
Preparing for an Outdoor Adventure. What to take and what to know before you go.

- Survival Equipment
- Celestrial Navigation
- Map & Compass Navigation
- Weather
- Poisonous Plants
- Venomous Snakes

Part 2:

Survival Tools and Techniques. What to do if something goes wrong?

- Basic Rules if You Are Lost
- Shelter
- How to Build a Campfire
- Useful Survival First Aid



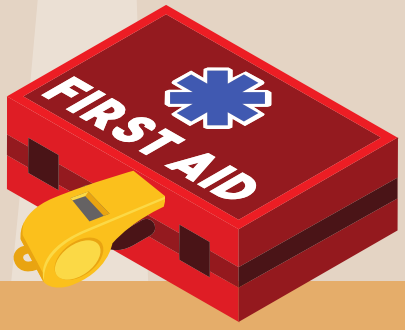
York County ENVIROTHON

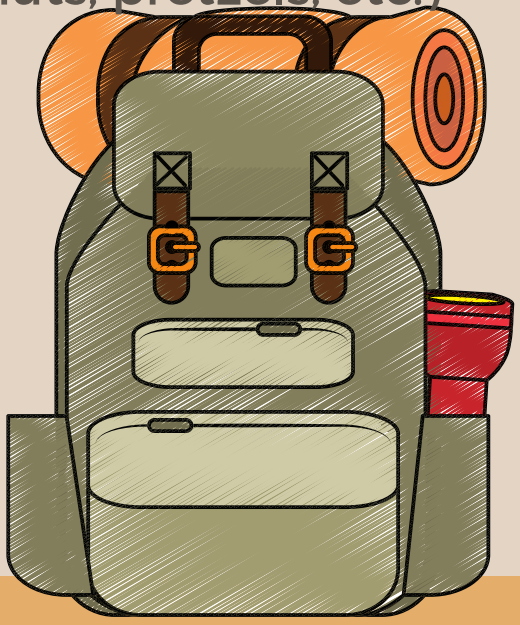
PART 1: PREPARING FOR AN OUTDOOR ADVENTURE

SURVIVAL EQUIPMENT

Outdoor safety isn't so much a game of survival AS IT IS AN EXERCISE IN PREPARATION. Know your limits, understand your environment, and prepare the right equipment.

- Identification and/or Medical Alert Tag or Bracelet (vital if unconscious when found)
- Brightly-colored Bandana (blaze orange is ideal)
- 1-2 Large Garbage Bags
- 1-2 Bottles or Canteens of Fresh Water
- Pocket Knife (sturdy, locking, folding knife is recommended)
- Fire Starter (lighter, matches, flint and steel)
- Portable Snacks (granola bars, nuts, pretzels, etc.)
- Loud Whistle
- Flashlight
- First Aid Kit
- Rope
- Compass
- Cell Phone
- Map or GPS





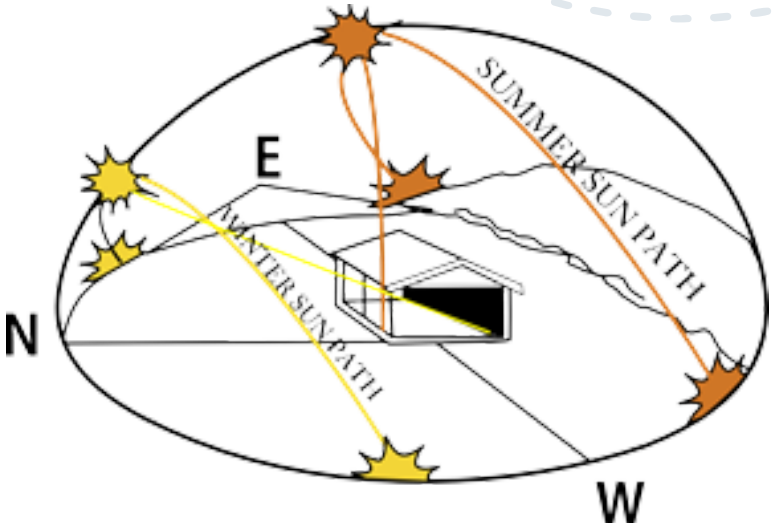
Note: In many wilderness areas, cell phone coverage may be inconsistent or unavailable. Additionally, leaf cover and geographic conditions may affect the use of a GPS unit. While both are good to have, please be aware that technological limitations may occur that can affect the use of a cell phone or GPS.

CELESTIAL NAVIGATION

Everyone that enjoys being outdoors should be able to use a map, compass, or GPS. In some survival situations, you may not have any of these. In these situations, we must turn to two things used by our ancestors – the Sun and the stars. The word "celestial" refers to anything pertaining to the sky or visible heaven. There are a number of ways to use the sun, moon and stars to find your way on the earth. The following are a few examples that may be employed easily in nature.

Using the sun & moon to find your way

To find direction when the Sun is shining, remember that the Sun rises in the East and sets in the West. At midday in the Northern Hemisphere, the Sun will be roughly South. Additionally, the moon rises in the east and sets in the west



Using the stars & night sky to find your way

The night sky can be used for orientation and navigation. This is done by first finding the North Star, or Polaris. Locate the Big Dipper in the northern sky. Depending upon the time of the year, constellation of stars may be tipped in different directions as it rotates around the polestar. As shown in the diagram, locate the two stars that form the outer edge of the Big Dipper. Draw an imaginary line straight through the two stars of the dipper edge and toward the Little Dipper. The line will point very close to the handle of the Little Dipper. The brightest star in the Little Dipper is at the end of its handle. This is the North Star, or Polaris, which is directly over the North Pole.

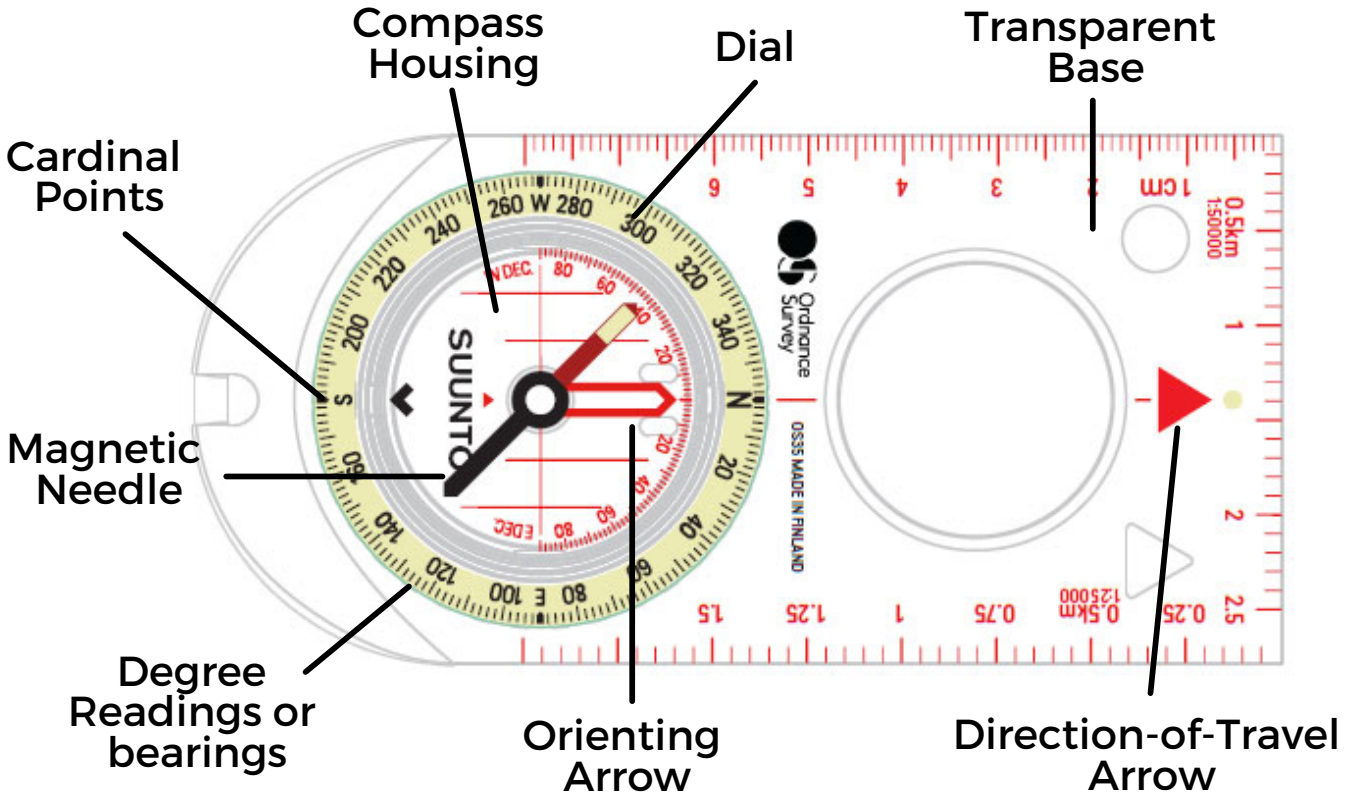


MAP & COMPASS NAVIGATION

No one should venture for a wilderness trip without a map and a compass. Being able to find direction by map and compass is a required skill for all wilderness travelers. Be sure to bring a map with you or obtain one at the park office. Maps that show topography, streams and other features are useful and recommended.

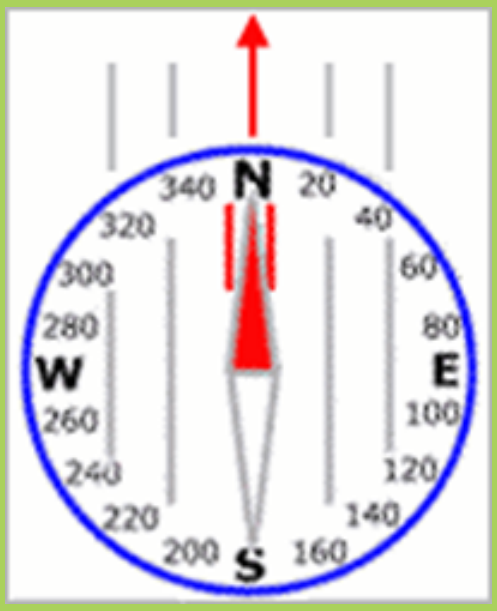
How a compass works: There is a huge magnetic field around the earth. It is huge, but not very strong. A compass has a small, magnetized needle inside the compass housing that floats in air, water or oil. The red end of the needle will always turn to point to magnetic north of the earth. When you know where north is, you can find any direction.

How to Use the Compass: The compass housing is surrounded by a dial, which is marked with the 360 degrees of a circle. The dial also is marked with the four cardinal points, north, south, east, and west. If you look at the dial on the compass housing and divide 360 by four (4), you will find each of the cardinal points at the following degree reading: North is at 0 degrees or 360 degrees, East is at 90 degrees, South is at 180 degrees and West is at 270 degrees.



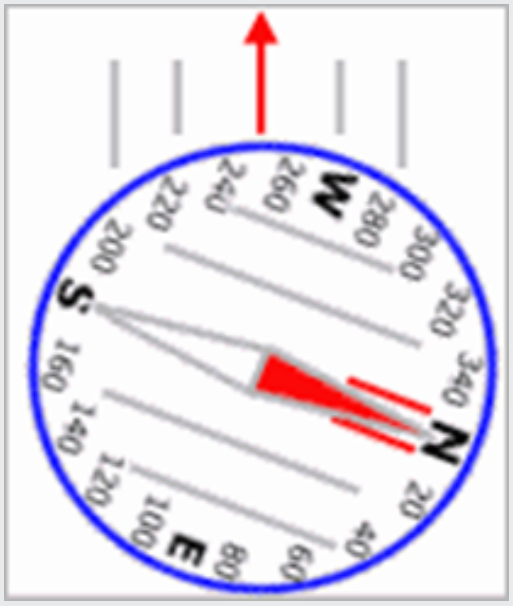
Finding North:

Hold the compass in front of you at waist height, with the direction-of-travel arrow pointing straight ahead. To find north, turn the compass housing until north is on the direction-of-travel arrow. Now, slowly turn yourself until the red end of the magnetic needle is pointing in the same direction as the orienting arrow and the direction-of-travel arrow. You are now facing north. Whenever you are facing north, the east is to your right, the west is to your left and south is behind you.



Finding the Direction of Travel:

Hold the compass in front of you at waist height, with the direction-of-travel arrow pointing in the direction you are traveling. Turn the compass housing until the orienting arrow is pointing in the same direction as the red end of the magnetic needle. Now, look at the degree reading which lines up with the direction-of-travel arrow and this is the direction in which you are traveling (remembering the degree readings described above).



WEATHER

When you are on your wilderness trip, some basic knowledge about weather will help you to take appropriate action to not get into trouble and risk your safety. It is always a good idea to check the weather forecast before heading out. Knowing what to expect can help you to be best prepared.

Advisory

An advisory is issued when a hazardous weather or hydrologic event is occurring, imminent or likely. Advisories are for less serious conditions than warnings that cause significant inconvenience and if caution is not exercised, could lead to situations that may threaten life or property.

Watch

A watch is used when the risk of a hazardous weather or hydrologic event has increased significantly, but its occurrence, location or timing is still uncertain. It is intended to provide enough lead-time so those who need to set their plans in motion can do so. A watch means that hazardous weather is possible. People should have a plan of action in case a storm threatens and they should listen for later information and possible warnings especially when planning travel or outdoor activities.

Warning

An advisory is issued when a hazardous weather or hydrologic event is occurring, imminent or likely. A warning means weather coniditions pose a threat to life or property. People in the path of the storm need to take protective action.

WATCH!
We have the ingredients for a cheesesteak!

WARNING!
Cheesesteaks on the table for dinner!



Source: National Oceanographic and Atmospheric Administration (NOAA)

POISONOUS PLANTS

Poison Ivy, Poison Oak, Poison Sumac

The most common ill effect on humans is a rash caused by contact with the plants or with their oil, called urushiol (yoo-ROO-she-ol). Rashes and blisters erupt on the skin, sometimes oozing and always itchy. The rash is NOT contagious. You must have contact with the oil in order for it to spread. Poison ivy and oak favor ground that has been disturbed and multiplies freely on embankments and along forest trails. Poison sumac, the largest plant of the three, contains the same allergen as poison ivy and oak and causes the same reaction, but grows predominantly in swamps. To prevent contact, do the following:

- Wear shoes and socks, not open sandals
- Look carefully at plants before walking through or touching them
- Be cautious when wiping sweat from your forehead with your arm; you may have had contact with urushiol on your sleeve, transferring it onto your face and in your eyes
- Wear loose clothing
- Be careful in water, plants growing at the water's edge can release enough urushiol into water to cause a reaction
- Wash pets that may have traveled through poison
- Decontaminate your belongings



Poison Ivy: The trademarks of this plant are its solid green, pointed leaves that hang from the stem in groups of three. It grows as both a vine and a shrub. The look of poison ivy can change with the seasons. It produces yellow-green flowers in the spring and its green leaves can change to yellow and red in autumn.



Poison Oak: Like its ivy counterpart, poison oak leaves also cluster in sets of three. The edges of the solid green leaves, while similar to an oak leaf, are less dramatic. Poison oak is most often seen in shrub form, but it can also grow as a vine.



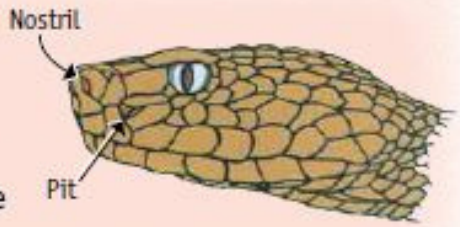
Poison Sumac: This rash-producer thrives in the water. It is usually found in swampy or boggy areas where it grows as small tree or tall shrub. Poison sumac leaves can have urushiol-filled black or brownish-black spots. The leaf stems contain seven to thirteen leaflets.

VENOMOUS SNAKES

Venomous Snakes in Pennsylvania

The first question most people ask when they see a snake is "Is that snake poisonous?" Technically, no snake is poisonous. A plant or animal that is poisonous is toxic when eaten or absorbed through the skin. A snake's venom is injected, so snakes are classified as venomous or nonvenomous.

In Pennsylvania, we have only three species of venomous snakes. They all belong to the pit viper family. A snake that is a pit viper has a deep pit on each side of its head that it uses to detect the warmth of nearby prey. This helps the snake locate food, especially when hunting in the darkness of night. These pits can be seen between the eyes and nostrils.



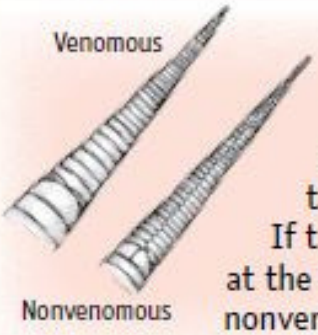
Pennsylvania Snakes Checklist

A complete list of Pennsylvania's *21 species* of snakes.

- ☐ Copperhead *(venomous)*
- ☐ Eastern Garter Snake
- ☐ Eastern Hognose Snake
- ☐ Eastern Massasauga *(venomous)*
- ☐ Eastern Milk Snake
- ☐ Eastern Rat Snake
- ☐ Eastern Ribbon Snake
- ☐ Eastern Smooth Earth Snake
- ☐ Eastern Worm Snake
- ☐ Kirtland's Snake
- ☐ Mountain Earth Snake
- ☐ Northern Black Racer
- ☐ Northern Brown Snake
- ☐ Northern Redbelly Snake
- ☐ Northern Ringneck Snake
- ☐ Northern Water Snake
- ☐ Queen Snake
- ☐ Rough Green Snake
- ☐ Shorthead Garter Snake
- ☐ Smooth Green Snake
- ☐ Timber Rattlesnake *(venomous)*

In Pennsylvania, all of our venomous snakes have slit-like pupils that are similar to a cat's eye. Nonvenomous snakes have round pupils, like a human eye.





If you find a shedded snake skin, look at the scales on the underside. If they are in a single row all the way to the tip of its tail, it came from a venomous snake. If the scales split into a double row at the tail, the shedded skin is from a nonvenomous snake.

Venomous snakes store their venom in a sac on each side of their head. This gives their head a triangular shape. However, some nonvenomous snakes can also flatten their heads, so using head shape alone is not the best way to identify a venomous snake.



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VENOMOUS SNAKES

The three species of venomous snakes in Pennsylvania are the timber rattlesnake, eastern massasauga and copperhead. Reports of venomous snakebites are rare in Pennsylvania. All three species are usually not aggressive and like to avoid humans by moving away or keeping still until the person is gone. **Never try to catch or handle a venomous snake.**



Timber Rattlesnake

Timber Rattlesnake

The timber rattlesnake has two different color phases: black and yellow. Both color phases have dark bands on their backs, a head with no markings and a dark tail. Their scales are deeply keeled.



Rattlesnakes add a new "rattle" to their tail each time they shed their skin. These rattles are loosely attached and can break off. The timber rattlesnake does not always rattle its tail before striking, especially when striking out of fear or defense.

Timber rattlesnakes like wooded hillsides with rocky outcrops. Timber rattlesnakes are a candidate species in Pennsylvania. This means that they are in danger of becoming a threatened species. Threatened species are in danger of becoming endangered. Endangered species are in danger of becoming extinct.

Eastern Massasauga

The eastern massasauga is a Pennsylvania endangered species and is the smallest of the Commonwealth's venomous snakes, averaging less than two feet in length. It is found in only a handful of locations within the western portion of the state. This snake lives only in wetland areas with dry fields or meadows next to it.



Eastern Massasauga

The eastern massasauga has a row of dark blotches running down the middle of its back with rows of smaller blotches along each side. Their scales are deeply keeled and the rattle is somewhat small.



Copperhead

Copperhead

The copperhead is the most common venomous snake in Pennsylvania.

It is tan or light brown, often with a tinge of pink. It has dark or reddish brown blotches on its back that are more narrow in the middle—like an hourglass. There are often small, dark spots in between the blotches. Its head is a copper color with no markings.

The scales are weakly keeled.

The copperhead is found in a variety of habitats.

PLAY

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For more information on snakes or snake publications, visit our website at www.fishandboat.com.

PART 2: SURVIVAL TOOLS AND TECHNIQUES

RULES IF YOU ARE LOST

1. Stop



Sit down and stay put until the fear, anger, and/or frustration has gone away

2. Think



THINK through your situation: What do you have that can help you in this situation? Your mind is your greatest survival tool!

3. Observe



Observe your surroundings: Where should you stay? If you told someone where you were going, people might be searching for you. Is there an open area where searchers would have a better chance of seeing you?

4. Plan



Plan your action: In most cases, the priority should be:

- find or make shelter
- build a fire for heat
- signal to attract attention
- find water



SHELTER

Building a survival shelter is an absolute priority if you are facing a survival situation in harsh or unpredictable weather. A good shelter must **protect you from the elements** and be **comfortable enough for resting and sleeping**. Most people cannot survive unprotected from rough weather for more than a couple of hours

Suitable Shelter Site	Unsuitable Shelter Site
• Dry, well drained, and reasonably flat • Comfortable distance from water • Close to firewood sources • Availability of building materials for shelter • Easily seen by search and rescue teams	• Too close to water (because of insects) • Dry riverbeds. Heavy rainfall can cause flash floods • Loose rocks, dead trees, or other natural growth that could fall onto the shelter • Low/High ground. Ravines or valleys could be damp and collect heavy cold air at night. Conversely, the tops of mountains are exposed to higher winds and other weather

If you have added a garbage bag, shelter tarp, an emergency blanket or an extra poncho to your gear, you are almost done with your shelter building. However, if you lack equipment, local conditions and materials will determine the type of wilderness shelter you build. Look for natural formations that could provide shelter. Examples include caves, rocky crevices and large trees with low-hanging limbs. (Remember, these may also be habitats for snakes, so be careful). If no natural formation is available to provide shelter, you will need to build your own. Make your shelter just large enough to accommodate you, especially in cold climates, because you are going to have to heat it. One of the best wilderness shelters is called the "debris hut." If you do not want to spend too much time and effort to build your shelter, the quickest type of wilderness shelter to build is the lean-to shelter.

SHELTER TYPES

The Cocoon

If it's almost dark and you can quickly collect dry debris (leaves, pine needles, bark) from the forest floor, make a pile two or three feet high and longer than you are tall. When you burrow into the pile, you are in a natural sleeping bag that protects against heat loss.





The Fallen Tree

The simplest shelter is a fallen tree that has enough room under it for you to crawl in. Lean branches against the windward side of the tree (so the wind is blowing into it and not against it) to make a wall. Make the wall thick enough to keep out wind. If you can build a fire on the open side of your shelter, the heat will help keep you warm.

The Lean-To

If you find a fallen tree without enough room under it, or a rock or a small overhang, you can build a simple lean-to. Start by leaning fallen limbs against the object, such as the top edge of an overhang, to create a wall. Lean the limbs at an angle to help shield rain. Cover the leaning limbs with leaves, boughs, pine needles, bark or whatever the forest offers. When you have built a thick wall, you can crawl underneath into your shelter. Remember to make your shelter no bigger than you need to fit you and anybody else with you.



HOW TO BUILD A CAMPFIRE

Step 1- The Right Spot

A safe fire site is one on which nothing will burn except the fuel you feed the flames. Fire cannot spread from this spot. Many camping areas have fire rings, grills, or stone fireplaces. Use those existing fire sites whenever you can. If an existing spot is not available, clear the area of debris, avoid overhanging branches and make sure your fire is a safe distance from tents and other flammable materials. One option is to build your fire on an earthen pad, but make sure to do it in a way that leaves no trace when you're done.

Step 2- The Right Material

The three types of natural fuel are tinder (material that burns easily such as dry grass or shavings whittled from a stick), kindling (slightly larger material such as twigs that will burn with a little encouragement) and firewood (dry branches that will be the main fuel of the fire). No shortcuts allowed — never use flammable liquids.



Step 3- The Right Shape

Two ways to build your fire are the tepee, in which you arrange the kindling in the shape of a tepee over the tinder, and the lean-to, in which you push a small stick in the ground at a 45-degree angle with the upper end pointing into the wind. Place the tinder beneath the stick and lean the kindling against it. Light the tinder and add kindling as needed. Add the larger branched last. Never leave a fire unattended.



Step 4- Put it OUT

Pour water on the fire, stir the ashes with a stick, pour some more. It's not out until you can run your bare hands through the coals.



USEFUL SURVIVAL FIRST AID

What is First Aid?

First aid is the first help given to the victim of an accident or other health emergency. If more attention is needed, first aid keeps an injured or ill person as safe as possible until trained medical personnel arrive. The goals of first aid include:

- Stop life threatening medical emergencies
- Protect person who is injured or ill from further harm
- Get the person under professional medical care

Wilderness Survival: First Aid and Health

When traveling into the wilderness, it is important to carry a complete first aid kit. If an accident happens in the wilderness, it is your responsibility to deal with the situation. The following sequence of actions to remember when dealing with situations should be followed every time:

1. **Remain calm.**
2. **Check the scene.** What caused the accident? Are there dangers? How many victims?
3. **Call for help.** Know the location, description of injury, time of injury, treatment received, and number of people.
4. **Approach safely.** Look for dangers. Tell the victim your name.
5. **Provide urgent treatment**. Is the person conscious and breathing? Is there severe bleeding? Is there evidence of pills, chemicals, or other poisons?
6. **Protect from further injury**. Avoid moving someone who is injured unless it is impossible to perform urgent first aid or the person is in a dangerous location.
7. **Treat every accident victim for shock**.
8. **Make a thorough examination.** Looks for other injuries that may be hidden and ask the victim questions if they are alert
9. **Plan a course of action**. If help is on the way, keep the victim comfortable. If help will be delayed, decide on a clear course of action.

Wilderness Survival: First Aid and Health, continued

Emergency	Treatment: 1st step, call for help
Shock Shock is a depression of all of the body processes and may follow any injury, regardless of how minor. Signs of shock include: • Restlessness or irritability • Confusion, fear, dizziness • Skin that is moist, clammy, cool and pale • A quick, weak pulse • Shallow, rapid and/or irregular breathing • Nausea and vomiting • Extreme thirst	• Try to eliminate the causes of shock by restoring breathing and circulation, controlling bleeding, relieving severe pain and treating wound • Monitor victim closely to make sure airway stays open for breathing • Help victim to lie down. If you do not suspect injuries to the back, neck or head, or fractures to the hip or leg, raise the feet about 12 inches to move blood from the legs to vital organs • Keep victim warm with blankets, coats or sleeping bags • Don't let the person eat or drink
Sudden Cardiac Arrest Sudden cardiac arrest is a leading cause of death. When a teen or adult has a sudden cardiac arrest, survival depends on immediately getting CPR from someone nearby.	According to the American Heart Association, Hands-Only CPR is CPR without mouth-to-mouth breaths. It is recommended for use by people who see a teen or adult suddenly collapse in an "out-of-hospital" setting (such as at home, at work or in a park). It consists of two easy steps: • Call 911 or send someone to do that • Push hard and fast in the center of the chest to the beat of "Staying Alive"
Bleeding	• Cover with sterile bandage or clean cloth • Use direct pressure to stop bleeding • Elevate injured body part above heart to slow blood flow • Use pressure at the pulse point between the injured area and the heart if bleeding fails to stop • After bleeding has been controlled, wash wounded area and apply a dressing or bandage
Fracture A fracture is classified as either simple (closed) or compound (open). Signs that a fracture is present include: • Pain in the affected area • Deformity in the area • The victim is unable to put weight on the area without pain • Swelling or a bluish color at the injury site	• Splint the joints above and below the fracture • Splint suspected fractures in the same position as you found them. Do not try to reposition an injured bone. • Be sure to pad the splints • Check the splint ties frequently to be sure they aren't hindering circulation or on the fracture • Cover all open wound with clean dressing before splinting • Treat for shock but do not raise a leg that may be broken • Apply an ice pack

Wilderness Survival: First Aid and Health, continued

Emergency	Treatment: 1st step, call for help
Concussions Concussions or other head injuries are often accompanied by leakage of watery blood from the nose or ears. Other symptoms may include: • Convulsions • Unresponsive pupils • Headache • Vomiting	• Keep the victim warm • Dispense pain reliever regularly • Allow time for the body to rest and repair; if victim were to fall asleep, wake periodically to check for signs of a concussion
Hypothermia Is when the body is losing more heat than it is able to produce, causing the core body temperature to drop. Hypothermia can quickly become life threatening. Signs of hypothermia: • Shivering uncontrollably • Feel cold and numb • Become tired and unable to think straight • Make poor decisions • Stumble, fall or lose consciousness	• Move person to a warm place • Monitor breathing and circulation • Remove any wet clothing • Warm the person slowly by wrapping blankets or by putting dry clothing on the person. Rapid warming may cause dangerous heart arrhythmias. Apply hot water bottles or hot packs that are first wrapped in towels or blankets. Warm the core first (trunk, abdomen) not the extremities (hands, feet). Warming hands and feet first can cause shock. • If fully conscious and able to swallow, have the person drink warm liquids.
Frostbite Is the freezing of a specific body part such as fingers, toes, earlobes, etc. Signs of frostbite: • Lack of feeling in the affected area • Skin that appears waxy, cold to the touch or is discolored (flushed, white or grey, yellow, or blue)	• Move person to a warm place • Handle the affected area gently, never rubbing • Gently warm affected by soaking in warm water (99-108 degrees F) until it appears red and feels warm • Avoid breaking any blisters • Do not allow to refreeze

Vocabulary

Celestial	Anything pertaining to the sky or visible heaven
Compound Fracture	A broken bone resulting in an open wound through which bone fragments usually protrude
First Aid	First help given to the victim of an accident or other health emergency
GPS	Global Positioning System, often the handheld varieties are used on outdoor adventures
Hydrologic	Pertaining to hydrology, the science dealing with the properties, distribution and circulation of water on and below the Earth's surface and in the atmosphere
Hypothermia	Subnormal temperature of the body
Poisonous	Full of or containing poison; a substance with an inherent property that tends to destroy life or impair health
Safety	The quality or state of being free from danger
Simple Fracture	Broken bone that does not result in an open wound
Shock	The depression of all the body process and may follow any injury, regardless of how minor
Survival	The condition of living or the state of being alive
Topography	The surface features of a place or region
Venomous	Having a gland for secreting venom; able to inflict a poisoned bite, sting or wound